INNOMATICS RESEARCH LABS

PROJECT REPORT

BookVerse Retail Business Intelligence & Sales Analytics

SQL-Based Operational Insights Using MySQL

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Technology	MySQL / SQL
Database	BookVerse_DB

INNOVATION. AUTOMATION. ANALYTICS

Executive Summary

This project analyzes an online bookstore scenario using a relational MySQL database named **BookVerse_DB**. The database contains three related tables—Books, Customers and Orders—and the analysis uses SQL queries to combine these tables and examine sales revenue, units sold, genre performance, customer spending, annual performance and inventory levels. The purpose is to convert operational sales records into information that can support business decisions.

The reported analysis covers 500 rows in each of the three tables. Across the analyzed orders, total revenue was **75,628.66**, units sold were **2,697**, and average order value was **151.26**. Romance generated the highest genre revenue at **13,086.98**, while Mystery recorded the highest unit sales at **504**. Kim Turner was identified as the highest-spending customer, with purchases totaling **1,398.90**. Revenue increased from 36,339.97 in 2023 to 36,775.33 in 2024, a reported increase of approximately 1.2%.

The findings suggest opportunities in stock replenishment, genre-based marketing, customer retention and regular sales monitoring. All monetary values are reproduced as supplied in the project material; no currency was specified.

Project Snapshot

Metric	Reported result

Database	BookVerse_DB
Tables	Books, Customers, Orders
Rows	500 in each table
Total revenue	75,628.66
Units sold	2,697
Average order value	151.26
Highest revenue genre	Romance – 13,086.98
Highest unit-sales genre	Mystery – 504 units
Highest-spending customer	Kim Turner – 1,398.90

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Note: The report follows the project content shown in the supplied BookVerse presentation screenshots and the project details provided in this conversation. It does not add unverified dataset fields or results.

1. Introduction

BookVerse is presented as an online bookstore that maintains information about books, customers and customer orders. As order and catalog information is stored across related tables, individual tables alone do not provide a complete view of business performance. SQL joins and aggregation functions can combine the data to answer practical questions about revenue, sales volume, customer value and product demand.

This project demonstrates a relational-database workflow: identify the business questions, understand the table structure, join related records, aggregate measures, interpret the output and recommend actions. The focus is SQL-based analysis rather than predictive modelling.

2. Business Scenario and Problem Statement

BookVerse wants to understand its sales performance and customer purchasing behaviour. Management needs to know which genres contribute most to revenue, which genres sell the most units, which customers spend the most, how revenue changes over time, and whether inventory levels require attention.

Problem statement: How can BookVerse use its books, customers and orders data to generate clear operational insights that support sales, marketing, customer-retention and inventory decisions?

3. Project Objectives

• Analyze overall book sales and revenue performance.

• Identify the best-performing genres by revenue and units sold.

• Understand customer spending and purchasing patterns.

• Evaluate monthly and yearly sales trends.

• Identify high-value customers.

• Review low-stock inventory and recommend replenishment actions.

4. Scope and Methodology

The project uses three relational tables in MySQL: Books, Customers and Orders. The analysis described in the presentation covers sales revenue, units sold, average order value, genre-level performance, customer spending, annual and monthly trends, and stock levels. The reported dataset size is 500 rows per table.

Methodology

• Database setup: create and use the BookVerse_DB database.

• Data loading: import the Books, Customers and Orders CSV datasets into their respective tables.

• Validation: check table row counts and inspect the loaded data.

• Relational analysis: use JOIN operations to connect orders to books and customers.

• Aggregation: apply SUM, COUNT, GROUP BY, ORDER BY and other SQL clauses to calculate businessmeasures.

• Interpretation: summarize the query results and convert them into business recommendations.

Tools and Technologies

Tool / technology	Purpose
MySQL	Relational database management and SQL analysis
MySQL Workbench	Database creation, data import and query execution
SQL	Joining, filtering, grouping and aggregating business data
CSV files	Source format for the Books, Customers and Orders datasets

The project material does not specify a separate ETL platform, cloud service or predictive analytics model; therefore these are not treated as implemented components in this report.

5. Database and Dataset Overview

The database is named **BookVerse_DB** and contains three tables. The presentation reports 500 rows in each table. The tables represent the product catalog, customer information and order transactions.

Table	Rows	Columns	Purpose
Books	500	7	Stores book details and inventory information.
Customers	500	3	Stores customer identity and country information.
Orders	500	6	Stores order date, customer, book, quantity and total amount.

Important fields

Books	Customers	Orders
Book_ID (key)	Customer_ID (key)	Order_ID (key)
Title	Name	Order_Date
Author	Country	Customer_ID (reference)
Genre		Book_ID (reference)
Published_Year		Quantity
Price		Total_Amount
Stock		

The table and field names above follow the project presentation. Primary-key and foreign-key labels are represented according to the ER structure shown there.

6. Database Relationships

The ER structure links Books and Customers to Orders. Each order references one book through Book_ID and one customer through Customer_ID. A book can be associated with multiple order records, and a customer can be associated with multiple order records. This structure enables analysis of sales by book attributes and customer attributes.

Relationship	Join condition	Analytical use
Books to Orders	Books.Book_ID = Orders.Book_ID	Analyze revenue and quantity by title, author or genre.
Customers to Orders	Customers.Customer_ID = Orders.Customer_ID	Analyze customer spend and order activity.

7. SQL Analysis and Findings

7.1 Overall sales and genre performance

The project groups orders by book genre and calculates total quantity and revenue. This helps compare genres using both sales volume and sales value.

```
SELECT b.Genre, SUM(o.Quantity) AS Total_Units_Sold, SUM(o.Total_Amount) AS Total_Revenue FROM
Books b JOIN Orders o ON b.Book_ID = o.Book_ID GROUP BY b.Genre ORDER BY Total_Revenue DESC;
```

Measure	Result
Total revenue	75,628.66
Total units sold	2,697
Highest revenue genre	Romance
Revenue from highest revenue genre	13,086.98
Highest units-sold genre	Mystery
Units sold in highest unit-sales genre	504

Interpretation: Romance contributes the largest reported revenue, while Mystery contributes the largest reported unit volume. Revenue leadership and unit-volume leadership are therefore not the same measure. BookVerse can use both measures when planning promotions and inventory.

7.2 Customer purchase analysis

Customer purchase analysis joins Customers with Orders, groups transactions by customer and calculates total spending and order count. The presentation identifies Kim Turner as the highest-spending customer, with a total spend of 1,398.90.

```
SELECT c.Customer_ID, c.Name, SUM(o.Total_Amount) AS Total_Spent, COUNT(o.Order_ID) AS Total_Orders
FROM Customers c JOIN Orders o ON c.Customer_ID = o.Customer_ID GROUP BY c.Customer_ID, c.Name
ORDER
BY Total_Spent DESC LIMIT 5;
```

Interpretation: Ranking customers by total spend helps identify high-value customers. Order frequency can provide additional context when designing loyalty offers or personalized recommendations. The report includes the top-spending customer stated in the presentation; the remaining top-five customer names and values were not supplied.

7.3 Monthly and yearly sales trends

The project groups order amounts by year and month to examine sales trends. The annual figures provided in the presentation are shown below.

```
SELECT YEAR(Order_Date) AS sales_year, MONTH(Order_Date) AS sales_month, SUM(Total_Amount) AS monthly_revenue FROM Orders GROUP BY YEAR(Order_Date), MONTH(Order_Date) ORDER BY sales_year, sales_month;
```

Year	Revenue	Change compared with prior year
2023	36,339.97	—
2024	36,775.33	+435.36 (about 1.2%)

Additional monthly findings reported in the presentation:

• Highest monthly revenue: May 2023 – 5,110.81.

• Lowest monthly revenue: December 2024 – 865.58.

• Average order value across the dataset: 151.26.

Interpretation: Revenue was slightly higher in 2024 than in 2023. Monthly variation suggests that BookVerse should review demand patterns regularly. The figures alone do not establish the causes of the monthly highs and lows.

8. Inventory and Stock Analysis

The Books table includes a Stock field. The project presentation reports that 5 books had zero stock and 37 books had stock levels of five or fewer. These records indicate items that may require inventory attention.

Inventory indicator	Reported count
Books with zero stock	5
Books with stock less than or equal to 5	37

Interpretation: Low-stock items should be reviewed alongside sales demand. A stock count alone does not show whether an item needs immediate replenishment; the business should also consider its sales rate, supplier lead time and reorder policy. Those additional measures were not provided in the project results.

9. Business Recommendations

The following recommendations are based on the reported SQL findings and are intended as practical actions for the BookVerse business.

9.1 Inventory planning

Review the 5 out-of-stock books and 37 books with stock levels of five or fewer. Prioritize replenishment after comparing each item's stock position with recent sales demand. Create a regular low-stock review process to reduce the risk of stockouts.

9.2 Genre-based marketing

Promote Romance titles to build on the genre's highest revenue contribution. Use Mystery-focused campaigns to support the genre's leading unit-sales volume. Measure campaign performance using revenue and units sold rather than relying on one metric.

9.3 Customer retention

Consider loyalty offers and personalized recommendations for high-value customers. Use both total spending and order frequency to segment customers, where the underlying order history supports those measures.

9.4 Sales monitoring

Track monthly and yearly revenue consistently. Investigate unusually high or low months using additional context such as promotions, stock availability and seasonal demand before drawing conclusions about the causes.

9.5 Business reporting

Maintain reusable SQL queries or a periodic report for revenue, units sold, average order value, genre performance and low-stock items. A dashboard can be added in a future phase if the business needs interactive reporting.

10. Conclusion

The BookVerse project demonstrates how MySQL and SQL can be used to organize and analyze retail bookstore data. By joining Books, Customers and Orders, the project evaluates sales revenue, units sold, average order value, customer spending, genre performance, yearly and monthly trends, and inventory levels. The reported results identify Romance as the highest-revenue genre, Mystery as the leading genre by units sold, and Kim Turner as the highest-spending customer. The annual revenue comparison shows a modest increase from 2023 to 2024. These findings can support better inventory planning, targeted marketing, customer retention and ongoing business reporting.

11. Limitations and Further Work

The report is based on the content and numerical results supplied in the BookVerse project presentation. The following details were not fully specified in the source material and have therefore not been invented:

- The currency for revenue and spending values.
- The complete list of genre-level revenue and quantity values.
- The full top-five customer ranking.
- The full monthly revenue series and the exact date coverage of the dataset.
- The detailed list of low-stock book titles.
- Data-cleaning rules, missing-value treatment and duplicate-record checks.

• Whether all orders represent completed and paid transactions.

Suggested future improvements

• Add data-quality checks for null values, duplicate IDs, invalid quantities and inconsistent dates.

• Create a complete genre-level summary table and customer ranking for reproducibility.

• Add reorder thresholds using sales velocity and supplier lead times.

• Build an interactive dashboard in Power BI if required for presentation or ongoing reporting.

• Document the date range, currency and business definitions for each KPI.

Appendix: Key SQL Concepts Used

SQL concept	How it is used in this project
INNER JOIN	Combines matching order records with book or customer records.
SUM()	Calculates total revenue and units sold.
COUNT()	Counts orders or records.
GROUP BY	Creates genre, customer, year and month summaries.
ORDER BY	Ranks results, such as highest revenue or customer spend.
LIMIT	Restricts the number of rows returned, such as the top five customers.
YEAR() / MONTH()	Extracts year and month from Order_Date for trend analysis.

End of Report